

Week 1 — Assignment 2

"Build Your Home Lab & Security Audit"

Due: Alongside the Weekend Assignment | Time estimate: 2-3 hours

What you need: Your own laptop/PC, this document, internet

Why This Assignment Exists

The first assignment was about capturing and analyzing traffic. This one is about understanding your own environment — the machine you use every day. A security professional who does not understand their own system cannot protect anyone else's.

You will:

1. Audit your own computer's network configuration
2. Check what is running and listening on your machine
3. Analyze your own network topology
4. Write a mini security assessment of your home network

Task 1 — Your Machine's Network Identity (30 min)

Step 1: Gather your network information

On Windows: Open Command Prompt and run: `ipconfig /all`

On Linux/Mac: Open Terminal and run: `ifconfig` or `ip a`

Step 2: Record the following

Item	Your value
Your computer's hostname	LAPTOP-41VHI60Q
Your IPv4 address	192.168.1.13
Your subnet mask	255.255.255.0
Your default gateway (router IP)	192.168.1.1
Your DNS server(s)	2401:4900:50:9::7dd, 2401:4900:50:9::8, 8.8.8.8, 8.8.4.4
Your MAC address	C0-BF-BE-19-B6-18
Are you on Wi-Fi or Ethernet?	Wi-Fi

```
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : 
Description . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
Physical Address. . . . . : C0-BF-BE-19-B6-18
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2401:4900:1c5e:ceec:b188:32ea:af68:9101(Preferred)
Temporary IPv6 Address. . . . . : 2401:4900:1c5e:ceec:99a7:cc6c:8a70:825e(Preferred)
Link-local IPv6 Address . . . . . : fe80::7196:96f8:ebcc:3fda%18(Preferred)
IPv4 Address. . . . . : 192.168.1.13(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 22 June 2026 10:13:12
Lease Expires . . . . . : 23 June 2026 12:48:46
Default Gateway . . . . . : fe80::1%18
                          192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 130072510
DHCPv6 Client DUID. . . . . : 00-01-00-01-2F-24-D9-BA-C0-BF-BE-19-B6-18
DNS Servers . . . . . : 2401:4900:50:9::7dd
                          2401:4900:50:9::8
                          8.8.8.8
                          8.8.4.4
Primary WINS Server . . . . . : 192.168.1.1
NetBIOS over Tcpip. . . . . : Enabled
```

Step 3: Answer these questions

1. What is your network's IP range?
192.168.1.0 – 192.168.1.255.
2. How many devices could potentially be on your network?
Up to **254 devices** can potentially exist on the network.
3. Is your DNS server your router or an external service?
The DNS service is primarily an **external service (Google DNS)** rather than the router.
4. Why does the MAC address matter? (Hint: it is a Layer 2 identifier — visible even before IP communication)
The MAC address uniquely identifies a device on a local network and allows network equipment to deliver data correctly at Layer 2 before IP communication occurs.

Task 2 — What Is Listening on Your Machine? (30 min)

Step 1: Check open ports

On Windows: `netstat -an`

On Linux/Mac: `ss -tlnp`

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING
TCP	0.0.0.0:445	0.0.0.0:0	LISTENING
TCP	0.0.0.0:5040	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49664	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49665	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49666	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49667	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49668	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49669	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49670	0.0.0.0:0	LISTENING
TCP	127.0.0.1:28385	0.0.0.0:0	LISTENING
TCP	127.0.0.1:28390	0.0.0.0:0	LISTENING
TCP	127.0.0.1:49350	0.0.0.0:0	LISTENING
TCP	127.0.0.1:49351	0.0.0.0:0	LISTENING
TCP	127.0.0.1:55092	127.0.0.1:55093	ESTABLISHED
TCP	127.0.0.1:55093	127.0.0.1:55092	ESTABLISHED
TCP	127.0.0.1:64023	127.0.0.1:49350	TIME_WAIT
TCP	127.0.0.1:64024	127.0.0.1:49350	TIME_WAIT
TCP	127.0.0.1:64025	127.0.0.1:49350	TIME_WAIT
TCP	127.0.0.1:64026	127.0.0.1:49350	TIME_WAIT
TCP	127.0.0.1:64027	127.0.0.1:49350	TIME_WAIT
TCP	127.0.0.1:64028	127.0.0.1:49350	TIME_WAIT
TCP	127.0.0.1:64029	127.0.0.1:49350	TIME_WAIT
TCP	127.0.0.1:64030	127.0.0.1:49350	TIME_WAIT
TCP	127.0.0.1:64031	127.0.0.1:49350	TIME_WAIT
TCP	127.0.0.1:64032	127.0.0.1:49350	TIME_WAIT
TCP	127.0.0.1:64034	127.0.0.1:49350	TIME_WAIT
TCP	127.0.0.1:64036	127.0.0.1:49350	TIME_WAIT
TCP	127.0.0.1:64037	127.0.0.1:49350	TIME_WAIT
TCP	127.0.0.1:64038	127.0.0.1:49350	TIME_WAIT
TCP	127.0.0.1:64039	127.0.0.1:49350	TIME_WAIT
TCP	127.0.0.1:64040	127.0.0.1:49350	TIME_WAIT
TCP	127.0.0.1:64041	127.0.0.1:49350	TIME_WAIT

Step 2: Record what you find

Local Address	Port	State	Service
0.0.0.0	135	LISTENING	Microsoft RPC
0.0.0.0	445	LISTENING	SMB File Sharing
0.0.0.0	5040	LISTENING	Windows Push Notifications
0.0.0.0	49664	LISTENING	Windows Dynamic RPC
0.0.0.0	49665	LISTENING	Windows Dynamic RPC
0.0.0.0	49666	LISTENING	Windows Dynamic RPC
0.0.0.0	49667	LISTENING	Windows Dynamic RPC
0.0.0.0	49668	LISTENING	Windows Dynamic RPC
0.0.0.0	49669	LISTENING	Windows Dynamic RPC
0.0.0.0	49670	LISTENING	Windows Dynamic RPC
127.0.0.1	28385	LISTENING	Local Application
127.0.0.1	28390	LISTENING	Local Application
127.0.0.1	49350	LISTENING	Local Service
127.0.0.1	49351	LISTENING	Local Service
192.168.1.13	139	LISTENING	NetBIOS Session Service

Step 3: Answer these questions

1. How many ports are in LISTENING state?
15 listening ports were identified.
2. Are any of these ports from the Common Ports list in your cheat sheet?
Yes, several common ports such as 135, 139, 445, 443 and 53 were identified.
3. Is port 3389 (RDP) open? If yes, what does that mean for security?
No. Port 3389 (Remote Desktop Protocol) is not open, reducing the risk of unauthorized remote access through RDP.
4. Is port 23 (Telnet) open? If yes, why is that a problem?
No. Telnet is not enabled. This is good because Telnet transmits usernames and passwords in plain text, making them vulnerable to interception.
5. What is the difference between 0.0.0.0:PORT and 127.0.0.1:PORT?
Ports bound to 0.0.0.0 are accessible from the network, while ports bound to 127.0.0.1 are restricted to the local computer only.

Step 4: Check running processes

On Windows: tasklist

On Linux/Mac: ps aux

List the top 10 processes you see. Identify any you do not recognize.

Process	Purpose
System Idle Process	Shows unused CPU resources
System	Core Windows kernel process
Secure System	Windows security features
Registry	Windows Registry management
smss.exe	Session Manager Subsystem
csrss.exe	Client/Server Runtime Subsystem
wininit.exe	Windows initialization process
services.exe	Service Control Manager
lsass.exe	Local Security Authority (authentication)
svchost.exe	Hosts Windows services

Task 3 — Scan Your Home Network (30 min)

Step 1: Find your network range

From Task 1, you know your IP and subnet mask. Calculate your network range.

192.168.1.0/24

Step 2: Ping sweep your network

On Windows (PowerShell):

```
1..254 | ForEach-Object { ping -n 1 -w 100 192.168.1.$_ | Select-String "Reply from" }
```

On Linux/Mac (Terminal):

```
for i in {1..254}; do ping -c 1 -W 1 192.168.1.$i | grep "bytes from"; done
```

Note: Replace 192.168.1 with YOUR network prefix. This scan is safe — ping only checks if a host responds.

Step 3: Record what you find

IP Address	Responded?	Likely Device
192.168.1.1	Yes	Router/Gateway
192.168.1.2	Yes	Connected device
192.168.1.4	Yes	Connected device
192.168.1.9	Yes	Connected device
192.168.1.11	Yes	Connected device
192.168.1.13	Yes	Your laptop/PC

Step 4: Answer these questions

1. How many devices responded to your ping sweep
6 devices responded.
2. Did you find any devices you did not expect?
No. All discovered devices appeared to belong to the home network.
3. If an attacker got onto your Wi-Fi, how many devices could they potentially reach?
An attacker could potentially reach all **6 active devices** and any additional devices that connect later.
4. What is the security risk of having all devices on one flat network?
A flat network allows all devices to communicate directly. If one device becomes compromised, an attacker may move laterally to other devices, access shared files, or spread malware.

Task 4 — Your Home Network Security Assessment (30 min)

Write a mini security assessment of your own home network. Use everything you learned this week. Write in your own words. Minimum 300 words.

My home network uses the private address range **192.168.1.0/24** with a subnet mask of **255.255.255.0**. My computer is connected through Wi-Fi and uses the IP address **192.168.1.13**. The default gateway is **192.168.1.1**, which provides internet connectivity to devices on the network. A ping sweep identified six active devices, including the router, my computer, and several other connected devices.

During the network audit, several listening ports were discovered. Common Windows networking ports such as **135 (RPC)**, **139 (NetBIOS)**, and **445 (SMB)** were active. Several dynamic RPC ports were also observed. These ports are used by Windows for communication and management tasks. No insecure services such as **Telnet (port 23)** were found. Additionally, **Remote Desktop Protocol (port 3389)** was not open, which reduces the risk of unauthorized remote access. Most active internet connections were using **HTTPS (port 443)**, indicating encrypted communication.

From a protocol security perspective, the network primarily relies on secure protocols. HTTPS protects web traffic through encryption, preventing attackers from easily reading transmitted data. DNS services are provided by external DNS servers, including Google's public DNS servers. No unencrypted protocols were observed during the audit.

CIA Triad Analysis

Confidentiality: Data confidentiality is protected through Wi-Fi authentication and encrypted HTTPS communications.

Integrity: Windows security features and Microsoft Defender help protect system files and data from unauthorized modification.

Availability: The network remains available through the router, internet connection, and Windows networking services. Malware or denial-of-service attacks could potentially affect availability.

AAA Framework Analysis

Authentication: Users authenticate through Windows user accounts and Wi-Fi credentials.

Authorization: Windows permissions determine what resources users can access.

Accounting: Windows Event Logs record user activity, security events, and system operations.

Structure your assessment like this:

1. Network Overview — What is your network range? How many devices? What types?
2. Open Ports & Services — What ports are open? Which services are running?
3. Protocol Security — What protocols does your network use? Any unencrypted protocols?
4. CIA Triad Analysis — Confidentiality, Integrity, Availability of your network
5. AAA Framework Analysis — Authentication, Authorization, Accounting on your network
6. Recommendations — List 5 specific improvements you could make

#	Improvement	Why it matters	How to do it
1	Use strong Wi-Fi passwords	Prevent unauthorized access	Use WPA2/WPA3 and a strong password
2	Keep Windows updated	Fix vulnerabilities	Install updates regularly
3	Disable unnecessary services	Reduce attack surface	Turn off unused sharing services
4	Enable firewall protection	Block unwanted connections	Keep Windows Defender Firewall enabled
5	Monitor connected devices	Detect unauthorized devices	Check router device list regularly

Task 5 — Bonus: Protocol Journal (optional, +10 bonus points)

For one full day, keep a protocol journal. Every time you use the internet, write down:

Time	What I did	Protocol used	Port	Encrypted?	What would an attacker see?
09:00	Checked email	HTTPS	443	Yes	Only traffic encrypted
09:15	Searched on Google	HTTPS	443	Yes	Search content hidden
09:30	Watched YouTube	HTTPS	443	Yes	Video content encrypted
10:00	Used Spotify	HTTPS	443	Yes	Music data encrypted
11:00	Opened Microsoft Word Online	HTTPS	443	Yes	Document contents protected
12:00	Checked Outlook email	HTTPS/IMAPS	443/993	Yes	Email contents hidden
14:00	Downloaded a pdf from a website	HTTPS	443	Yes	Download encrypted
15:00	Used ChatGPT	HTTPS	443	Yes	Conversation encrypted
17:00	Played Roblox	HTTPS/TCP	443	Yes	Game traffic encrypted
19:00	Browsed news websites	HTTPS	443	Yes	Website content encrypted

Fill in at least 10 entries. This teaches you to think like a security analyst in your daily life.

How to Submit

Create a document with:

1. Task 1: Your network information table + answers
2. Task 2: Your port scan table + answers + process list
3. Task 3: Your ping sweep results + answers
4. Task 4: Your written security assessment (300+ words)
5. Task 5 (bonus): Your protocol journal (10+ entries)

Name it: Week1_HomeLab_Audit_YourName.md (or .txt or .pdf)

Grading Rubric

Criteria	Points	Your score
Task 1: Network identity recorded correctly	/15	
Task 1: Subnetting questions answered	/10	
Criteria	Points	Your score
Task 2: Port scan completed and analyzed	/15	
Task 2: Process identification	/5	
Task 3: Ping sweep completed	/10	
Task 3: Security reasoning about flat networks	/10	
Task 4: Written assessment quality (300+ words)	/20	
Task 4: CIA + AAA connections accurate	/10	
Task 4: 5 practical recommendations	/5	
Bonus: Protocol journal (10+ entries)	+10	
Total	/100 (+10 bonus)	

Tips

- Do this assignment on your HOME network, not a public Wi-Fi.
- Take screenshots of your ipconfig, netstat, and ping sweep output.
- Be honest in your assessment. If your Wi-Fi has no password, write that down.
- The best security assessments are the ones that make you uncomfortable.